

Motivational Letter

After graduating from high school, I became a mentor for young people at a boarding school, where I experienced first-hand the distress and challenges many young people go through during their adolescent years. Ever since, I have had a passion for young people and a desire to help them, and I have grown a heart especially for the ones suffering from emotional disorders and mental illness in general.

In recent years it has become clear that adolescents are experiencing an ongoing decline in mental health. The emerging fields of computational psychiatry and translational neuroscience have the potential to deepen our understanding of the causes and mechanisms behind emotional disorders, bridging the gap between the molecular, psychosomatic, and cognitive processes – exactly what is needed to address this most pressing issue of our time. I feel a deep vocation to aid in this endeavour, for which the MSc in Neural Systems and Computation program is well suited, as it combines a broad interdisciplinary field, frontier research, and a collaborative environment.

As part of a ‘perception and action’ course, a classmate, Samuel Nehrer, and I created a package in the Julia programming language for simulation and model fitting with active inference. This fills a gap in the computational psychiatry community, as it offers active inference cognitive modelling in an easily accessible software, which only requires minimal coding expertise. Through this project I gained knowledge on Bayesian inference, model fitting and collaborative software development, for which I am very grateful.

I find the path that I am currently treading both fascinating and fulfilling and seek to continue down this path. The research conducted at UZH and ETH Zürich stands out in its field as among the most premier research institutions, and in particular the research conducted by the TNU has caught my attention. The goal of understanding neural systems through an interdisciplinary approach and preparing students for a career conducting independent and meaningful research aligns tremendously with my own aspirations. By joining the MSc in Neural Systems and Computation I hope to lay a strong foundation for a career specializing in computational approaches to mental health and neuroscience. Contributing with a passion for young people as well as a theoretical foundation in neuroscience and computational psychiatry I look forward to collaborating with peers and members of the faculty on projects expanding our understanding of human cognition.